



Importance of Climate Change Awareness for Engineers

Introduction

Climate change awareness is the understanding of how human activities, especially the burning of fossil fuels, deforestation, industrial production, and unsustainable resource use, are changing Earth's climate system. For engineers, climate change awareness is not only an environmental topic; it is a professional necessity because engineers design, build, operate, and maintain the systems on which society depends.

Engineers work on buildings, roads, bridges, dams, power plants, machines, transport systems, communication networks, water supply systems, software, manufacturing processes, and energy technologies. Many of these systems either contribute to greenhouse gas emissions or are affected by climate-related risks such as heatwaves, floods, droughts, cyclones, sea-level rise, and extreme rainfall.

The World Meteorological Organization reported that 2024 was the warmest year in the 175-year observational record, with global mean temperature about $1.55 \pm 0.13^\circ\text{C}$ above the 1850–1900 average. It also highlighted major social and economic disruption from extreme weather, record ocean heat, and sea-level rise.

Meaning of Climate Change Awareness in Engineering

For engineering students, climate change awareness means knowing:

1. The causes of climate change.
2. The impact of climate change on society, infrastructure, industry, and natural resources.
3. The role of engineering in reducing greenhouse gas emissions.
4. The need to design climate-resilient systems.
5. The ethical responsibility of engineers toward present and future generations.

Climate change awareness helps engineers move from traditional design thinking to sustainable and resilient design thinking. Instead of asking only, “Will this structure or machine work today?”, engineers must also ask, “Will it remain safe, efficient, affordable, and useful under future climate conditions?”

Why Climate Change Awareness Is Important for Engineers

Engineers design long-life infrastructure

Many engineering projects have long service lives. Bridges, dams, highways, power stations, buildings, water systems, ports, and railway networks may operate for 50 to 100 years. The American Society of Civil Engineers notes that many infrastructure systems have service lives of 50 to 100 years and must remain functional, durable, and safe during that period.

This means an engineer designing a bridge today must consider not only present rainfall, temperature, wind, and flood levels, but also how these conditions may change in the future. Climate change awareness helps engineers design systems that do not fail prematurely.

Engineering projects can increase or reduce emissions

Engineering decisions strongly affect greenhouse gas emissions. The International Energy Agency reported that total energy-related CO₂ emissions reached a record 37.8 gigatonnes in 2024.

Engineers influence emissions through choices such as:

- Type of energy used.
- Materials selected.
- Manufacturing processes.
- Transport systems.
- Building design.
- Waste management.
- Energy efficiency.
- Product life cycle.
- Maintenance and end-of-life disposal.

For example, a civil engineer choosing low-carbon concrete, a mechanical engineer improving motor efficiency, an electrical engineer designing renewable energy systems, and a software engineer reducing data-centre energy consumption all contribute to climate action.

Climate change affects safety and reliability

Extreme weather events can damage infrastructure and disrupt services. Floods can damage bridges and roads. Heatwaves can reduce power plant efficiency and cause railway tracks to expand. Cyclones can damage coastal structures. Droughts can reduce hydropower generation and water supply.

The IPCC's Working Group II report assesses climate change impacts on ecosystems, biodiversity, human communities, vulnerabilities, and adaptation capacity, including cities, settlements, water, health, and key infrastructure.

An engineer who is unaware of climate risks may design systems based on outdated assumptions. This can lead to unsafe structures, service failure, higher repair costs, and risk to human life.

Engineers are problem-solvers for sustainable development

Engineers are not only technical workers; they are problem-solvers for society. The World Federation of Engineering Organizations connects engineering work with Sustainable Development Goals such as sustainable cities, responsible consumption, climate action, clean water, clean energy, and resilient infrastructure.

Climate-aware engineers can create solutions such as:

- Renewable energy systems.
- Energy-efficient buildings.
- Electric mobility.
- Smart grids.
- Flood-resistant drainage.
- Sustainable transport.
- Green manufacturing.
- Waste-to-energy systems.
- Carbon capture technologies.
- Climate-resilient agriculture technologies.
- Water conservation systems.

Role of Engineers in Climate Change Mitigation

Mitigation means reducing the causes of climate change, mainly by lowering greenhouse gas emissions and increasing carbon removal.

Energy-efficient design

Engineers can reduce energy consumption by designing efficient machines, motors, pumps, compressors, HVAC systems, lighting, buildings, and industrial processes.

Examples:

- Use high-efficiency motors.
- Improve insulation in buildings.
- Recover waste heat from industries.
- Use LED lighting and smart controls.
- Design efficient public transport systems.
- Reduce friction and losses in mechanical systems.

Renewable energy development

Electrical, mechanical, civil, and electronics engineers play important roles in renewable energy technologies such as solar, wind, hydro, biomass, geothermal, and green hydrogen.

Engineering tasks include:

- Designing solar PV systems.
- Developing wind turbine components.
- Improving battery storage.
- Designing smart grids.
- Integrating renewable energy into power systems.
- Improving power electronics and inverters

Sustainable materials

Civil, mechanical, chemical, and materials engineers can reduce emissions by selecting materials with lower environmental impact.

Examples:

- Low-carbon cement and concrete.
- Recycled steel and aluminium.
- Fly ash or slag-based construction materials.
- Bamboo, engineered timber, and local materials where suitable.
- Recyclable polymers and biodegradable materials.
- Lightweight materials for vehicles.

Circular economy

The circular economy aims to reduce waste by designing products and systems for reuse, repair, recycling, and longer life.

Engineers can apply circular economy principles by:

- Designing products that are easy to repair.
- Reducing material waste during manufacturing.
- Reusing industrial by-products.
- Designing modular components.
- Planning for recycling at the end of product life.

The Engineering Council advises engineers to consider the full life cycle of products, processes, and systems, use resources efficiently, apply life cycle assessment, and promote circularity.

Role of Engineers in Climate Change Adaptation

Adaptation means adjusting engineering systems so that society can live safely under changing climate conditions.

Climate-resilient infrastructure

Climate-resilient infrastructure is designed to withstand future climate stresses such as floods, extreme heat, high winds, and sea-level rise.

Examples:

- Stronger bridges in flood-prone areas.
- Higher drainage capacity in cities.
- Heat-resistant road materials.
- Coastal protection structures.
- Flood barriers and retention ponds.
- Elevated electrical equipment in flood zones.
- Cyclone-resistant housing.

ASCE states that there is an immediate need for engineers to include resilience to future climate changes in project design criteria, and that many existing codes and standards are not structured to fully address climate adaptation.

Water resource management

Climate change affects rainfall patterns, groundwater recharge, river flow, and drought frequency. Engineers must design better water systems.

Important engineering solutions include:

- Rainwater harvesting.
- Wastewater recycling.
- Smart irrigation.
- Leak detection in water networks.
- Desalination where appropriate.
- Flood forecasting systems.
- Stormwater management.
- Watershed protection.

Urban planning and smart cities

Cities are highly vulnerable to climate change because they concentrate people, buildings, transport, industries, and energy use.

Climate-aware urban engineering includes:

- Green roofs.
- Urban forests.
- Cool roofs and reflective pavements.
- Efficient public transport.
- Walkable and cycle-friendly streets.
- Flood-safe zoning.
- Decentralized waste management.

- Energy-efficient buildings.
- Smart sensors for air quality, heat, and water levels.

Disaster risk reduction

Engineers help reduce disaster risks through early warning systems, safe buildings, emergency shelters, resilient roads, and reliable communication networks.

Examples:

- Seismic and cyclone-resistant structures.
- Flood mapping.
- Real-time monitoring of dams and bridges.
- Emergency power systems.
- Communication networks during disasters.
- Evacuation route planning.

Importance for Different Engineering Branches

Civil Engineering

Civil engineers design buildings, roads, bridges, dams, tunnels, airports, railways, ports, and drainage systems. Climate change awareness helps them design infrastructure that can withstand floods, heatwaves, storms, sea-level rise, and changing soil conditions.

Key areas:

- Green buildings.
- Low-carbon construction materials.
- Flood-resistant design.
- Sustainable urban drainage.
- Climate-resilient transport infrastructure.
- Life cycle assessment of buildings.

Mechanical Engineering

Mechanical engineers work with machines, engines, HVAC systems, manufacturing equipment, and energy systems. Climate awareness helps them improve efficiency and reduce emissions.

Key areas:

- Efficient engines and turbines.
- Renewable energy machines.
- Heat recovery systems.
- Low-energy cooling systems.
- Electric vehicle components.
- Sustainable manufacturing.

Electrical and Electronics Engineering

Electrical engineers are central to clean energy transition and power system reliability.

Key areas:

- Solar and wind power integration.
- Smart grids.
- Energy storage.
- Electric vehicle charging.
- Efficient motors.
- Power electronics.

- Microgrids for rural and disaster-prone areas.

Computer Science and IT Engineering

Digital technologies also consume energy, especially data centres, AI systems, blockchain systems, and cloud computing. Climate-aware computer engineers can reduce the environmental footprint of digital systems.

Key areas:

- Energy-efficient algorithms.
- Green data centres.
- Smart energy management.
- Climate modelling.
- IoT-based environmental monitoring.
- AI for flood prediction and energy optimization.

Chemical Engineering

Chemical engineers are important in industries such as fertilizers, cement, petroleum, pharmaceuticals, plastics, and materials.

Key areas:

- Cleaner production.
- Carbon capture.
- Green hydrogen.
- Biofuels.
- Waste-to-energy.
- Low-emission industrial processes.
- Pollution control.

Automobile Engineering

Transport is a major source of emissions. Automobile engineers can support climate action through cleaner mobility.

Key areas:

- Electric vehicles.
- Hybrid vehicles.
- Lightweight vehicle design.
- Efficient engines.
- Battery technology.
- Hydrogen fuel cells.
- Public transport systems.

Ethical Responsibility of Engineers

Engineering has a strong ethical dimension. Engineers must protect public safety, health, welfare, and the environment. The Engineering Council states that engineers and technicians should apply professional and responsible judgement, use systems thinking, keep sustainability knowledge up to date, and provide options and solutions that support sustainable development.

Ethical climate-aware engineering means:

- Avoiding designs that cause unnecessary environmental damage.
- Considering future generations.
- Being honest about climate risks.
- Not ignoring scientific evidence.

- Designing for safety, equity, and affordability.
- Informing clients and decision-makers about sustainable alternatives.
- Balancing economic, environmental, and social impacts.

Climate Change Awareness and Engineering Education

Engineering students should learn climate change awareness because they are future designers, innovators, managers, researchers, and policymakers.

Important learning areas include:

1. Basics of climate science.
2. Carbon footprint calculation.
3. Life cycle assessment.
4. Renewable energy systems.
5. Sustainable materials.
6. Environmental impact assessment.
7. Disaster-resilient design.
8. Green building concepts.
9. Waste management.
10. Environmental laws and standards.
11. Sustainable development goals.
12. Ethics and professional responsibility.

Climate awareness should not be limited to environmental engineering students. It is relevant to every engineering branch.

Benefits of Climate Change Awareness for Engineers

Better design decisions

Climate-aware engineers make better choices in materials, site selection, energy systems, drainage, insulation, transport planning, and manufacturing.

Reduced project risk

Projects designed without climate awareness may face damage, repair costs, legal issues, insurance problems, and public criticism. Climate-resilient design reduces long-term risk.

Innovation and career opportunities

Climate change creates demand for engineers in renewable energy, electric vehicles, green buildings, sustainable construction, environmental consulting, climate data analysis, smart grids, and disaster management.

Cost savings

Energy-efficient and resource-efficient designs can reduce operating costs over the life of a project.

Public safety

Climate-aware design protects people from floods, heat stress, storms, water shortages, and infrastructure failure.

Contribution to national development

Countries need climate-aware engineers to build sustainable cities, reliable power systems, clean industries, resilient agriculture, efficient transport, and safe infrastructure.

Indian Context

India is rapidly developing infrastructure, industry, housing, transport, and energy systems. This makes climate-aware engineering especially important. The IEA reported that India's energy-related CO₂ emissions rose by 5.3% in 2024, driven by rapid economic growth, infrastructure development, rising energy demand, and prolonged heatwaves that increased electricity consumption.

For Indian engineers, important climate-related challenges include:

- Urban flooding.
- Heatwaves.
- Water scarcity.
- Coastal erosion.
- Air pollution.
- High cooling demand.
- Renewable energy integration.
- Sustainable construction.
- Climate-resilient agriculture infrastructure.
- Public transport and electric mobility.

Engineering students in India can contribute through solar energy, smart irrigation, low-cost cooling, flood-resistant drainage, affordable green buildings, waste management, and rural electrification.

Examples of Climate-Aware Engineering Solutions

Example 1: Green building

A green building uses natural lighting, ventilation, insulation, efficient cooling, solar panels, rainwater harvesting, and low-carbon materials. It reduces energy use and improves occupant comfort.

Example 2: Climate-resilient bridge

A bridge in a flood-prone area can be designed with higher clearance, stronger foundations, better drainage, and flood-resistant materials.

Example 3: Smart irrigation

Sensors can measure soil moisture and provide water only when needed. This saves water and energy.

Example 4: Solar microgrid

A solar microgrid can provide electricity to rural areas and maintain power supply during disasters.

Example 5: Electric public transport

Electric buses and metro systems reduce urban air pollution and greenhouse gas emissions, especially when powered by renewable electricity.

Key Principles for Engineers

Engineering students should remember the following principles:

1. Design for the future, not only the present.
2. Reduce energy and material consumption.
3. Use renewable energy wherever possible.
4. Consider the full life cycle of a product or project.
5. Design for repair, reuse, and recycling.
6. Include climate risks in safety calculations.
7. Use data, modelling, and scientific evidence.
8. Think in systems, not isolated components.
9. Protect vulnerable communities.
10. Balance development with environmental responsibility.

Challenges Faced by Engineers

Climate-aware engineering is important, but it also has challenges:

- Lack of updated climate data.
- Higher initial cost of sustainable materials.
- Resistance from clients or industries.
- Inadequate building codes or standards.
- Lack of awareness among professionals.
- Uncertainty in future climate projections.
- Short-term profit thinking.
- Limited access to advanced technologies.

However, these challenges can be managed through research, education, government policy, professional ethics, innovation, and interdisciplinary teamwork.

Climate change awareness is essential for engineers because engineering decisions shape the future of society. Engineers design the systems that produce energy, transport people, manufacture goods, manage water, build cities, and protect communities. These systems must be low-carbon, efficient, resilient, safe, and sustainable.

A climate-aware engineer is not only technically skilled but also socially responsible. Such an engineer understands that every design has environmental, economic, and social consequences. By applying climate awareness, engineers can reduce emissions, protect infrastructure, improve public safety, support sustainable development, and contribute to a better future for humanity.

Weather and Climate

Weather and climate are two closely related but different concepts in the study of the Earth's atmosphere. Weather refers to the short-term condition of the atmosphere at a particular place and time. It includes daily changes in temperature, humidity, rainfall, wind speed, cloud cover, and air pressure. Weather can change very quickly, sometimes within a few hours or from one day to another. For example, a day may begin sunny, become cloudy in the afternoon, and end with rainfall in the evening. Therefore, weather is temporary, local, and highly variable.

Climate, on the other hand, refers to the long-term average pattern of weather conditions in a particular region. It is usually studied over a period of 30 years or more. Climate helps us

understand the general atmospheric conditions of a place, such as whether a region is usually hot, cold, dry, humid, or rainy. Unlike weather, climate does not change quickly. It develops over a long period and influences natural systems, agriculture, human settlements, and economic activities. For example, the climate of Rajasthan is generally hot and dry, while the climate of coastal Kerala is warm and humid.

Difference Between Weather and Climate

The main difference between weather and climate lies in their time scale. Weather describes short-term atmospheric changes, while climate explains long-term atmospheric patterns. Weather is observed for hours, days, or weeks, whereas climate is studied over decades or centuries. Weather is mostly local and affects immediate daily activities such as travel, farming, outdoor work, and event planning. Climate has a broader influence because it affects agriculture, biodiversity, water resources, urban planning, housing patterns, and the overall lifestyle of people in a region.

Weather is influenced by immediate atmospheric conditions such as air pressure, temperature, moisture, wind movement, local topography, water bodies, and seasonal changes. Climate is influenced by long-term factors such as latitude, altitude, distance from the sea, ocean currents, mountain ranges, vegetation, and global atmospheric circulation. Human activities such as deforestation, industrialization, urbanization, and the burning of fossil fuels are also affecting climate patterns across the world.

Characteristics of Weather

The characteristics of weather include its short duration, frequent changes, and local nature. Weather conditions are measured through instruments such as thermometers, barometers, anemometers, rain gauges, and hygrometers. Weather forecasting helps people prepare for rain, storms, heatwaves, cold waves, and cyclones. However, weather predictions are reliable only for a short period because atmospheric conditions change rapidly.

Characteristics of Climate

The characteristics of climate include long-term stability, regional influence, and predictable patterns. Climate is studied through long-term records of temperature, rainfall, wind, sunshine, and humidity. It helps scientists identify seasonal trends, rainfall distribution, drought-prone areas, and long-term temperature changes. Climate also helps in understanding major global issues such as global warming, sea-level rise, melting glaciers, and changing rainfall patterns.

Significance of Weather and Climate

Weather and climate are significant because they affect almost every aspect of life on Earth. In agriculture, weather determines daily crop growth, irrigation needs, and harvesting schedules, while climate decides which crops are suitable for a region. For example, rice grows well in warm and wet climates, while wheat grows better in cooler conditions. In human health, extreme weather events such as heatwaves, storms, and cold waves can directly affect people's safety, while long-term climate change can increase health risks such as heat stress, waterborne diseases, and the spread of vector-borne diseases.

Weather and climate also influence economic activities. Weather affects daily operations in transport, aviation, shipping, construction, tourism, and fishing. Climate plays an important role in long-term planning for industries, agriculture, energy production, and infrastructure

development. Renewable energy sources such as solar, wind, and hydropower depend heavily on weather conditions and long-term climate patterns. For example, solar power generation depends on sunlight, wind power depends on wind speed, and hydropower depends on rainfall and river flow.

Impact on Ecosystems and Biodiversity

The natural environment is also shaped by weather and climate. Climate determines the type of forests, grasslands, deserts, and wildlife found in a region. Short-term weather changes affect plant growth, animal migration, breeding cycles, and water availability. Ecosystems and biodiversity depend on stable climate conditions, but rapid climate change can disturb natural balance and threaten many species.

Role in Disaster Preparedness

Weather forecasting and climate studies are important for disaster preparedness. Weather forecasts help governments and people take immediate action during cyclones, floods, thunderstorms, heatwaves, and heavy rainfall. Climate studies help in long-term planning, such as building flood-resistant infrastructure, improving drainage systems, protecting coastal areas, and managing water resources. Therefore, both weather and climate knowledge are essential for safety, planning, and sustainable development.

Challenges Related to Weather and Climate

Weather and climate present many challenges. One of the biggest challenges is climate change, which is caused mainly by increasing greenhouse gas emissions from industries, vehicles, power plants, and deforestation. Climate change is altering long-term temperature and rainfall patterns and increasing the frequency and intensity of extreme weather events. Heatwaves, droughts, floods, cyclones, and heavy rainfall events are becoming more severe in many parts of the world.

Urbanization and Land Use Changes

Urbanization is another major challenge. Rapid growth of cities creates urban heat islands, where cities become hotter than surrounding rural areas due to concrete buildings, roads, vehicles, and reduced green cover. Deforestation and land-use changes also disturb natural climate regulation by reducing the ability of forests to absorb carbon dioxide and maintain rainfall patterns. These human activities increase environmental stress and make communities more vulnerable to weather and climate-related disasters.

Resource Scarcity and Sea-Level Rise

Resource scarcity is also linked to changing weather and climate patterns. Irregular rainfall, droughts, and rising temperatures affect water supply, food production, and energy security. In many regions, climate-related stress can lead to economic losses, migration, and conflicts over resources. Rising sea levels caused by melting ice and warming oceans threaten coastal areas, damage ecosystems, and displace human populations.

Data Collection and Forecasting Limitations

Another challenge is the limitation of data collection and forecasting. Although modern technology has improved weather prediction and climate modeling, atmospheric systems are complex and difficult to predict with complete accuracy. Many developing regions still lack

proper weather stations, climate data, and early warning systems. This makes disaster preparedness and long-term planning more difficult.

Public Awareness and Policy Gaps

Public awareness and policy implementation are also major challenges. Many people still do not fully understand the difference between weather and climate or the seriousness of climate change. Weak environmental policies, poor planning, and lack of international cooperation can slow down climate action. Developing countries often face greater risks because they have fewer resources to adapt to climate change, even though they may contribute less to global emissions.

Weather and climate are both important parts of the Earth's atmospheric system, but they differ in meaning, duration, and impact. Weather refers to short-term atmospheric conditions, while climate refers to long-term weather patterns. Weather affects daily life, while climate shapes agriculture, ecosystems, economy, settlements, and long-term development. Understanding the difference between weather and climate is essential for environmental awareness, disaster management, sustainable planning, and climate change adaptation. As climate-related challenges increase, awareness of weather and climate becomes necessary for students, policymakers, engineers, farmers, and society as a whole.

Overview of Global Climate System

The global climate system is a complex and interactive system that controls the climate of the Earth. It includes the atmosphere, oceans, land surface, ice and snow cover, and living organisms. These parts do not work separately; they continuously exchange heat, water, gases, and energy. The climate system is important because it determines weather patterns, rainfall, temperature, winds, ocean circulation, ecosystems, and long-term climate change. According to the attached file, understanding the climate system is necessary to understand, detect, and predict natural as well as human-induced climate change.

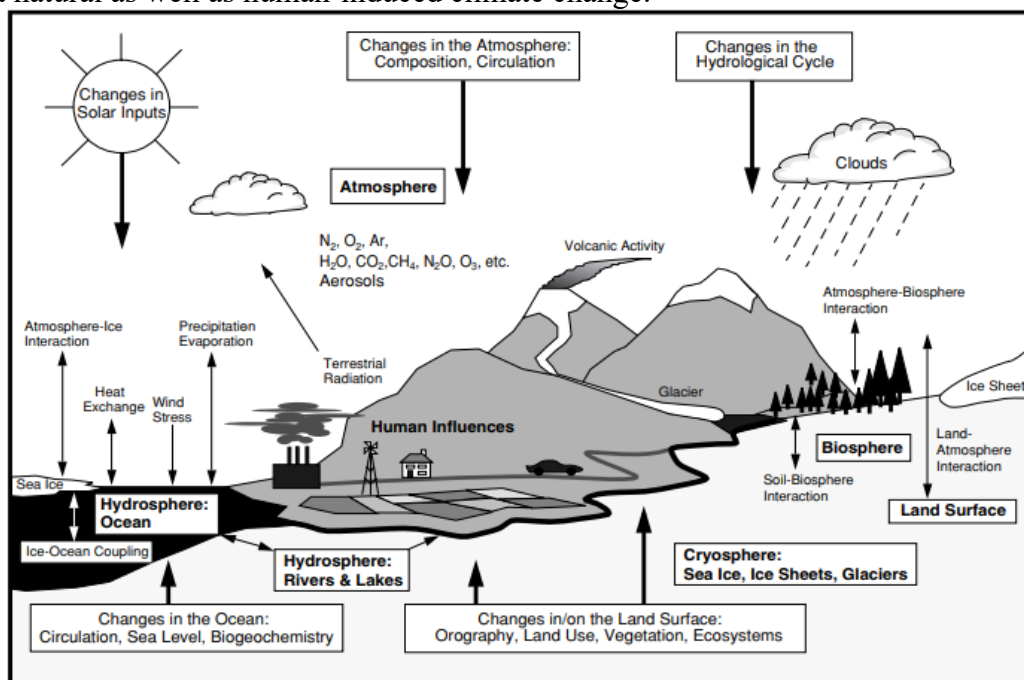


Figure 1: Schematic view of the components of the global climate system (bold), their processes and interactions (thin arrows) and some aspects that may change (bold arrows).

Climate Variables

The main climate variables are temperature, rainfall, humidity, wind, cloud cover, solar radiation, and atmospheric pressure. These variables directly affect human life, agriculture, water resources, health, and ecosystems. However, climate is not controlled only by surface weather conditions. It is also influenced by the vertical structure of the atmosphere, ocean currents, land surface features, ice cover, vegetation, soil moisture, atmospheric composition, solar radiation, and volcanic activity.

Major Components of the Global Climate System

The global climate system has five major components: atmosphere, hydrosphere, cryosphere, land surface, and biosphere. The file explains that these components are influenced by external forcing mechanisms, especially solar energy from the Sun. Human activities are also considered an external forcing because they can change atmospheric composition and land surface conditions.

Atmosphere

The atmosphere is the most unstable and rapidly changing part of the climate system. It is mainly composed of nitrogen, oxygen, and argon. These major gases do not strongly interact with infrared radiation. However, trace gases such as carbon dioxide, methane, nitrous oxide, ozone, and water vapour absorb and emit infrared radiation. These gases are called greenhouse gases and play an essential role in the Earth's energy balance.

Water vapour is especially important because it is a strong natural greenhouse gas and exists in different forms such as vapour, cloud droplets, and ice crystals. Clouds and aerosols also influence climate by interacting with incoming solar radiation and outgoing infrared radiation. Therefore, the atmosphere controls temperature, rainfall, storms, winds, and the greenhouse effect.

Hydrosphere

The hydrosphere includes all liquid water on and below the Earth's surface. It includes oceans, seas, rivers, lakes, groundwater, and aquifers. Oceans are the most important part of the hydrosphere because they cover about 70% of the Earth's surface. They store and transport large amounts of heat and carbon dioxide.

Ocean circulation is driven by winds, temperature differences, and salinity differences. This circulation is slower than atmospheric circulation, but it has a major influence on climate. Because oceans have large thermal inertia, they reduce sudden temperature changes and act as a regulator of the Earth's climate.

Cryosphere

The cryosphere includes ice sheets, glaciers, snow fields, sea ice, and permafrost. It is important because ice and snow have high albedo, meaning they reflect a large amount of solar radiation back into space. This helps regulate the Earth's temperature.

The cryosphere also affects ocean circulation and sea level. Ice sheets store huge amounts of water, so changes in their volume can cause sea-level changes. Melting glaciers and ice sheets can therefore influence coastal regions, ocean circulation, and the global water cycle.

Land Surface

The land surface includes soil, vegetation, mountains, deserts, forests, and human-modified land. Land surface conditions control how solar energy is absorbed, reflected, or returned to the atmosphere. Vegetation and soil moisture affect evaporation and surface temperature. For example, moist soil uses energy for evaporation, which can reduce surface heating.

The roughness of the land surface, caused by topography and vegetation, affects wind movement near the ground. Land surfaces can also release dust into the atmosphere, which interacts with radiation and influences climate. Human activities such as agriculture, deforestation, irrigation, and urbanization can change the land surface and affect climate.

Biosphere

The biosphere includes all living organisms on land and in the oceans. It plays a major role in the carbon cycle and the exchange of greenhouse gases. Plants absorb carbon dioxide during photosynthesis and store carbon in vegetation and soils. Forests and marine plants are especially important carbon sinks.

The biosphere also releases gases such as methane, nitrous oxide, and volatile organic compounds. These gases can affect atmospheric chemistry, aerosol formation, and climate. Climate also affects the biosphere, and evidence of past climate is preserved in fossils, tree rings, pollen, and other biological records.

Interactions Among Climate System Components

The components of the climate system are connected through exchanges of mass, heat, momentum, water, and gases. The atmosphere and oceans exchange heat and water vapour through evaporation. This forms part of the hydrological cycle, which includes evaporation, condensation, cloud formation, rainfall, runoff, and return flow to oceans.

The atmosphere and oceans also exchange carbon dioxide. Cold polar waters absorb carbon dioxide and sink into the deep ocean, while warmer upwelling waters release carbon dioxide near the surface. Sea ice affects the exchange of heat and gases between the ocean and atmosphere. The biosphere affects carbon dioxide levels through photosynthesis and respiration, and vegetation affects water vapour transfer through evapotranspiration.

The Figure 1, shows the main components of the global climate system and their interactions, including atmosphere–ocean interaction, atmosphere–ice interaction, land–atmosphere interaction, biosphere interaction, volcanic activity, solar input, and human influences.

Solar Energy and Global Energy Balance

The Sun is the ultimate source of energy for the climate system. Solar radiation drives atmospheric circulation, ocean currents, evaporation, cloud formation, rainfall, and biological productivity. The text states that each square metre of the Earth's spherical surface receives an average of 342 W/m² of solar radiation outside the atmosphere. About 31% is reflected back into space by clouds, the atmosphere, and the Earth's surface. The remaining energy is absorbed by the atmosphere, land, and oceans.

For a stable climate, incoming solar radiation must be balanced by outgoing radiation emitted by the Earth system. The Earth returns heat to the atmosphere through infrared radiation, sensible heat, and water vapour. This energy exchange helps maintain the present global mean surface temperature of about 14°C , according to the file.

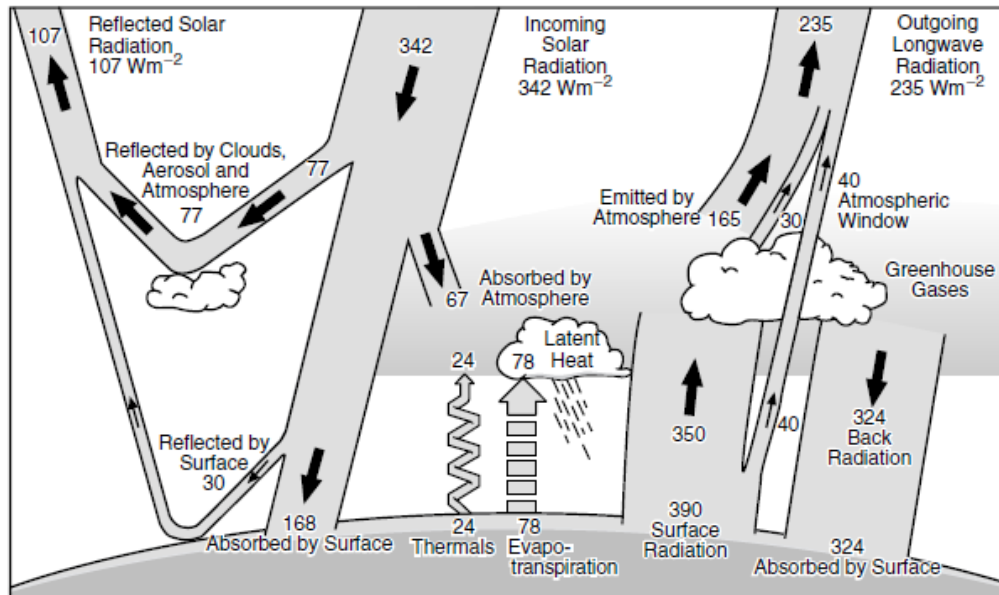


Figure 1.2: The Earth's annual and global mean energy balance. Of the incoming solar radiation, 49% (168 Wm^{-2}) is absorbed by the surface. That heat is returned to the atmosphere as sensible heat, as evapotranspiration (latent heat) and as thermal infrared radiation. Most of this radiation is absorbed by the atmosphere, which in turn emits radiation both up and down. The radiation lost to space comes from cloud tops and atmospheric regions much colder than the surface. This causes a greenhouse effect. Source: Kiehl and Trenberth, 1997: Earth's Annual Global Mean Energy Budget, Bull. Am. Met. Soc. 78, 197-208.

Natural Greenhouse Effect

The natural greenhouse effect is an essential part of the Earth's energy balance. Greenhouse gases absorb infrared radiation emitted by the Earth's surface and atmosphere. They then emit radiation in all directions, including back toward the Earth's surface. This keeps the lower atmosphere and surface warmer than they would be without greenhouse gases.

The file explains that without considering the greenhouse effect, the Earth would radiate energy to space at an effective temperature of about -19°C , while the actual average surface temperature is about 14°C . This difference shows the importance of natural greenhouse gases in maintaining conditions suitable for life. The energy balance diagram on page 6, Figure 1.2, illustrates incoming solar radiation, reflected radiation, outgoing longwave radiation, surface radiation, back radiation, thermals, latent heat, and greenhouse gases.

Radiative Forcing

Radiative forcing refers to a change in the balance between incoming solar radiation and outgoing infrared radiation. If the Earth receives more energy than it emits, positive radiative forcing occurs and the climate tends to warm. If the Earth loses more energy than it receives, negative radiative forcing occurs and the climate tends to cool.

Radiative forcing may be caused by natural factors such as changes in solar radiation or volcanic aerosols. It may also be caused by human activities such as greenhouse gas emissions,

aerosol emissions, and land-use change. The climate system responds to radiative forcing in order to restore energy balance.

Natural Climate Variability

Climate varies naturally on different time scales. Some variations are caused by external factors such as solar changes, volcanic eruptions, and changes in Earth's orbit. Other variations arise from internal interactions among the atmosphere, oceans, land, ice, and biosphere.

Different parts of the climate system respond at different speeds. The troposphere responds within days or weeks, the stratosphere within months, and the oceans over decades to centuries or even millennia because of their large heat capacity. The biosphere may respond quickly to droughts but slowly to long-term climate change.

ENSO, NAO, and Regional Climate Variability

Regional climate can be more variable than global climate. This is because changes in one region may be balanced by opposite changes elsewhere. Large-scale climate patterns often arise from interactions between atmospheric circulation, oceans, and land surfaces.

One important example is the El Niño–Southern Oscillation, or ENSO, which results from interaction between the atmosphere and ocean in the tropical Pacific. El Niño and La Niña events influence weather and climate across the world. Another example is the North Atlantic Oscillation, or NAO, which affects the climate of Europe and parts of Asia, especially in winter.

Feedbacks in the Climate System

Feedbacks make the climate system more complex. A feedback occurs when the result of a process affects the original process. A positive feedback strengthens the original change, while a negative feedback reduces it.

Water vapour feedback is an example of positive feedback. When the Earth warms, the atmosphere can hold more water vapour. Since water vapour is a greenhouse gas, this can further increase warming. Radiative damping is an example of negative feedback. When temperature increases, the Earth emits more infrared radiation, which helps limit further warming. Cloud feedbacks and carbon cycle feedbacks are also important but complex.

Extreme Events

Extreme events are weather events that are far from average conditions, such as heat waves, droughts, floods, and heavy rainfall. Climate is connected with the probability distribution of weather events. Even a small change in average climate can change the frequency or intensity of extreme events. Societies and ecosystems are often more vulnerable to extremes than to average conditions because extreme events can cause sudden damage to health, agriculture, infrastructure, and biodiversity.

Human Influence on the Climate System

Human influence on climate became significant after the Industrial Revolution. Activities such as burning fossil fuels, biomass burning, deforestation, agriculture, urbanization, and industrial production release greenhouse gases and aerosols into the atmosphere. These activities change atmospheric composition and land surface properties.

The file explains that carbon dioxide, methane, nitrous oxide, ozone, chlorofluorocarbons, and aerosols affect radiative forcing. Land-use change affects albedo, surface roughness, evaporation, and greenhouse gas exchange. Urbanization can create an urban heat island, where cities become warmer than surrounding rural areas due to buildings, roads, reduced vegetation, and human heat emissions.

Enhanced Greenhouse Effect

The enhanced greenhouse effect occurs when human activities increase the concentration of greenhouse gases in the atmosphere. Higher greenhouse gas concentrations increase the absorption and emission of infrared radiation. This reduces the amount of heat escaping to space and creates positive radiative forcing.

As a result, the surface and lower atmosphere tend to warm. The file notes that if carbon dioxide were doubled instantly, outgoing infrared radiation would be reduced, producing radiative forcing. However, the actual climate response is complex because of feedbacks involving water vapour, clouds, oceans, land, and the biosphere.

Role of Aerosols

Aerosols are tiny solid or liquid particles in the atmosphere. They may come from natural sources such as dust and volcanic eruptions, or from human activities such as industry, energy use, and land-use change. Aerosols can scatter sunlight back into space, causing cooling. Some aerosols, such as soot, absorb solar radiation and cause atmospheric heating.

Aerosols also affect clouds by changing the number, size, and properties of cloud droplets. This can influence cloud brightness, rainfall formation, and climate. Their effects are complicated because aerosols have short atmospheric lifetimes and vary strongly from region to region.

Climate Models and Projections

Climate models are tools used to study the climate system and project future climate change. They are based on physical, chemical, and biological principles. General Circulation Models represent the atmosphere and oceans using mathematical equations. Atmosphere–Ocean General Circulation Models combine atmosphere and ocean models with other components such as land surface and cryosphere.

Because the climate system is highly complex and non-linear, models require powerful computers. They are used to simulate present climate, study past climate, and project future climate under different greenhouse gas and aerosol scenarios. The file explains that projections are not exact predictions because future population, technology, economy, and energy use are uncertain. Therefore, scientists use scenarios to explore possible future climate outcomes.

Observing the Climate System

Climate observation is necessary to detect and understand climate change. Observations include land and ocean surface temperature, sea surface temperature, rainfall, wind, atmospheric composition, ocean data, ice cover, and satellite measurements. Instrumental observations became more widespread from the mid-19th century, while upper-air observations expanded from the late 1950s.

Satellites now provide global observations of many climate system components, but surface and in-situ observations are still needed for accuracy. Data must be corrected for changes in instruments, station locations, measurement techniques, urbanization effects, and satellite biases. Data assimilation systems combine observations with models to produce a more complete picture of the climate system.

Importance of Understanding the Global Climate System

Understanding the global climate system is important because it helps scientists and policymakers explain past climate variations, monitor present climate change, and project possible future changes. It also helps in disaster preparedness, agriculture, water management, energy planning, biodiversity conservation, and infrastructure development.

For engineers and planners, knowledge of the climate system is useful in designing climate-resilient buildings, roads, bridges, drainage systems, water supply systems, renewable energy systems, and coastal protection structures. Since climate change affects rainfall, temperature, floods, droughts, storms, sea level, and ecosystems, climate system knowledge is essential for sustainable development.

The global climate system is an interconnected system made up of the atmosphere, hydrosphere, cryosphere, land surface, and biosphere. It is driven mainly by solar energy and regulated by the exchange of heat, water, gases, and radiation among its components. Natural factors such as solar variability, volcanic eruptions, ocean circulation, and internal climate patterns cause climate variability. Human activities such as fossil fuel burning, deforestation, urbanization, and industrial emissions are now major forces affecting the climate system.

A clear understanding of the climate system helps us explain weather and climate, understand the greenhouse effect, study climate variability, detect climate change, and prepare for future climate risks. Therefore, the overview of the global climate system is a basic but essential topic in environmental science, geography, engineering, and climate studies.

Greenhouse Effect

The greenhouse effect is a natural process that helps maintain the temperature of the Earth. The term “greenhouse effect” is derived from a greenhouse, which is a glass house used for growing plants. In a greenhouse, sunlight enters through the glass and warms the plants and air inside. However, the heat produced inside does not escape easily, so the temperature inside the greenhouse remains warm. This warmth is useful for the proper growth of plants.

A similar process takes place in the Earth’s atmosphere. During the day, solar radiation from the Sun reaches the Earth and warms its surface. At night, when the Earth cools, heat is radiated back from the surface toward the atmosphere. Some of this heat escapes into space, but a part of it is absorbed by greenhouse gases present in the atmosphere. These gases trap heat and keep the Earth’s surface warm. This natural greenhouse effect makes life possible on Earth by preventing the planet from becoming too cold.

However, due to human activities, the concentration of greenhouse gases in the atmosphere has increased. This has strengthened the greenhouse effect and caused a rise in the average temperature of the Earth. This rise in temperature is known as global warming and has many harmful environmental effects.

The greenhouse effect is the process by which greenhouse gases in the Earth's atmosphere absorb infrared radiation emitted from the Earth's surface and prevent some of the heat from escaping into space. This process keeps the Earth's surface warm and helps maintain conditions suitable for life.

Greenhouse Gases

Greenhouse gases are gases that absorb infrared radiation and contribute to the greenhouse effect. These gases allow sunlight to enter the atmosphere but trap some of the heat radiated back from the Earth's surface.

The main greenhouse gases include carbon dioxide, methane, nitrous oxide, water vapour, ozone, and chlorofluorocarbons. Carbon dioxide is released mainly by the burning of fossil fuels, deforestation, and industrial activities. Methane is released from agriculture, livestock, wetlands, and landfills. Nitrous oxide is released from fertilizers and certain industrial processes. Chlorofluorocarbons, also known as CFCs, are human-made gases that were earlier used in refrigerators, air conditioners, and aerosol sprays.

The major sources of greenhouse gases are factories, automobiles, power plants, deforestation, agriculture, and waste disposal sites. As the number of industries, vehicles, and human activities increases, the amount of greenhouse gases in the atmosphere also increases. These gases trap more heat, raise the Earth's temperature, and contribute to global warming.

Causes of Greenhouse Effect

Burning of Fossil Fuels

Burning fossil fuels is one of the major causes of the enhanced greenhouse effect. Fossil fuels such as coal, petroleum, and natural gas are widely used for transportation, electricity generation, industries, and domestic purposes. When these fuels are burned, they release large amounts of carbon dioxide into the atmosphere.

With the increase in population, industrialization, and transportation, the use of fossil fuels has increased rapidly. As a result, more carbon dioxide is released into the atmosphere, strengthening the greenhouse effect and increasing global temperature.

Deforestation

Deforestation is another important cause of the greenhouse effect. Trees and plants absorb carbon dioxide from the atmosphere during photosynthesis and release oxygen. When forests are cut down, the number of trees available to absorb carbon dioxide decreases.

In addition, when trees are burned or left to decay, the carbon stored in them is released back into the atmosphere as carbon dioxide. Therefore, deforestation increases the concentration of greenhouse gases and contributes to global warming.

Farming and Agricultural Activities

Agricultural activities also contribute to the greenhouse effect. The use of nitrogen-based fertilizers releases nitrous oxide, which is a powerful greenhouse gas. Paddy fields, livestock, and organic waste also release methane into the atmosphere.

As the demand for food increases, farming activities expand, leading to greater emissions of methane and nitrous oxide. These gases trap heat and increase the warming of the atmosphere.

Industrial Waste and Landfills

Industries and factories release harmful gases into the atmosphere through the burning of fuels and chemical processes. These emissions include carbon dioxide, methane, nitrous oxide, and other pollutants.

Landfills also contribute to the greenhouse effect. When organic waste decomposes in landfills, it produces methane and carbon dioxide. Methane is especially harmful because it has a strong heat-trapping ability. Therefore, improper waste management adds to greenhouse gas emissions.

Effects of Greenhouse Effect

Global Warming

Global warming is one of the major effects of the enhanced greenhouse effect. It refers to the gradual increase in the average temperature of the Earth's atmosphere and surface. The main cause of global warming is the increased concentration of greenhouse gases such as carbon dioxide, methane, and nitrous oxide.

These gases are mainly released by the burning of fossil fuels, vehicle emissions, industrial activities, deforestation, and agriculture. Global warming can lead to melting glaciers, rising sea levels, changing rainfall patterns, droughts, floods, heatwaves, and loss of biodiversity.

Depletion of Ozone Layer

The ozone layer is found in the upper part of the atmosphere called the stratosphere. It protects the Earth from harmful ultraviolet radiation from the Sun. Depletion of the ozone layer allows more ultraviolet rays to reach the Earth's surface. This can cause skin cancer, eye problems, damage to crops, and harm to marine life.

Ozone layer depletion is mainly caused by chlorofluorocarbons and other ozone-depleting substances. Although ozone depletion and the greenhouse effect are different environmental problems, some gases such as CFCs are connected with both because they can damage the ozone layer and also act as greenhouse gases.

Smog and Air Pollution

Smog is a form of air pollution formed by the combination of smoke, fog, and harmful gases. It may be caused by natural processes, but human activities are the main reason for its increase. Vehicle emissions, industrial smoke, burning of fossil fuels, agricultural fires, and forest fires release pollutants such as nitrogen oxides, sulphur oxides, and particulate matter.

These pollutants react in the atmosphere and form smog. Smog reduces visibility, causes breathing problems, irritates the eyes, and harms plants and animals. It is a serious problem in many urban and industrial areas.

Acidification of Water Bodies

The increase in air pollution and greenhouse gas emissions can also affect water bodies. Pollutants such as sulphur dioxide and nitrogen oxides mix with rainwater and form acid rain.

When acid rain falls into rivers, lakes, ponds, and streams, it increases the acidity of water bodies.

Acidification of water bodies harms aquatic plants and animals. It can disturb the natural balance of aquatic ecosystems and reduce biodiversity. Rainwater may also carry pollutants from the atmosphere and land into water bodies, further degrading water quality.

The greenhouse effect is a natural and essential process that keeps the Earth warm enough to support life. Without it, the Earth would be too cold for most living organisms. However, human activities have increased the concentration of greenhouse gases in the atmosphere, causing an enhanced greenhouse effect. This has led to global warming, climate change, air pollution, ozone-related problems, and damage to ecosystems.

To reduce the harmful effects of the enhanced greenhouse effect, it is necessary to control the burning of fossil fuels, reduce deforestation, promote renewable energy, improve waste management, use public transport, plant more trees, and adopt sustainable agricultural and industrial practices. Public awareness and responsible environmental action are essential for protecting the Earth's climate system.

Greenhouse Gas

Greenhouse gas emissions refer to the release of gases into the atmosphere that trap heat and contribute to global warming. These gases allow sunlight to enter the Earth's atmosphere but absorb and retain some of the heat radiated back from the Earth's surface. As a result, the temperature of the Earth gradually increases. This process is known as the enhanced greenhouse effect.

The main greenhouse gases responsible for global warming are carbon dioxide, methane, nitrous oxide, and fluorinated gases. Each gas has different sources, different heat-trapping capacity, and different effects on the environment. Understanding these gases is important for developing effective methods to reduce their impact on climate change.

Types of Greenhouse Gases

Greenhouse gases are directly linked with global warming. Each type of greenhouse gas has specific sources and different levels of impact on the climate. Knowing these differences helps governments, industries, and individuals plan better strategies for reducing emissions.

Carbon Dioxide

Carbon dioxide is one of the most important greenhouse gases. It is mainly released by the burning of fossil fuels such as coal, petroleum, and natural gas. These fuels are used in electricity generation, industries, transportation, and domestic activities.

Carbon dioxide remains in the atmosphere for a very long time, sometimes for hundreds of years. Because of this, it plays a major role in long-term climate change. Deforestation also increases carbon dioxide levels because fewer trees are available to absorb carbon dioxide from the atmosphere.

Methane

Methane is another powerful greenhouse gas. It is released during oil and natural gas production, coal mining, livestock farming, rice cultivation, and the decomposition of organic waste in landfills. In livestock farming, methane is produced during the digestive process of animals such as cattle.

Although methane stays in the atmosphere for a shorter time than carbon dioxide, it is much more effective at trapping heat. Therefore, even small increases in methane concentration can have a strong warming effect.

Nitrous Oxide

Nitrous oxide is mainly released from agricultural activities, especially through the use of synthetic and organic fertilizers. It is also emitted from certain industrial processes, fuel combustion, and waste treatment.

Nitrous oxide has a much stronger warming effect than carbon dioxide. It also contributes to the depletion of the stratospheric ozone layer, which protects the Earth from harmful ultraviolet radiation.

Fluorinated Gases

Fluorinated gases are human-made gases used in industries and products such as refrigerators, air conditioners, solvents, foam-blowing agents, and certain manufacturing processes. These gases include hydrofluorocarbons, perfluorocarbons, sulphur hexafluoride, and nitrogen trifluoride.

Fluorinated gases are present in smaller quantities compared to carbon dioxide, but they are extremely powerful greenhouse gases. Some of them can remain in the atmosphere for thousands of years and have a very high heat-trapping capacity.

Sources of Greenhouse Gas Emissions

Greenhouse gas emissions mainly come from human activities. The burning of fossil fuels is one of the largest sources of emissions. Coal, oil, and natural gas are used to produce electricity, run vehicles, power industries, and heat buildings. These activities release large amounts of carbon dioxide into the atmosphere.

Industries and factories are also major contributors. Manufacturing processes, cement production, chemical industries, and energy-intensive production facilities release carbon dioxide, methane, nitrous oxide, and fluorinated gases.

Transportation is another important source of greenhouse gas emissions. Cars, buses, trucks, ships, trains, and airplanes burn fossil fuels and release carbon dioxide and other pollutants. As the number of vehicles increases, emissions from the transport sector also rise.

Agriculture and land use also contribute significantly to greenhouse gas emissions. Livestock farming releases methane, while the use of fertilizers releases nitrous oxide. Deforestation, land clearing, and poor soil management reduce the Earth's ability to absorb carbon dioxide and increase emissions.

Waste disposal sites and landfills release methane and carbon dioxide when organic waste decomposes. Poor waste management increases greenhouse gas emissions and also creates air, soil, and water pollution.

Effects of Greenhouse Gas Emissions on the Environment

The effects of greenhouse gas emissions are serious and wide-ranging. One of the major effects is global warming, which causes a gradual rise in the average temperature of the Earth. This rise in temperature affects weather patterns, rainfall, oceans, ecosystems, agriculture, and human health.

Rising sea levels are another major effect. As global temperatures increase, glaciers and polar ice caps melt, adding more water to the oceans. Warmer ocean water also expands, further increasing sea levels. This threatens coastal areas, islands, and low-lying regions.

Greenhouse gas emissions also contribute to ocean acidification. When carbon dioxide is absorbed by oceans, it changes the chemistry of seawater and makes it more acidic. This harms marine life, especially corals, shellfish, and other organisms that depend on calcium carbonate for their shells and skeletons.

Wildlife is also affected by climate change. Changes in temperature, rainfall, and habitats force many species to migrate. Some species may not be able to adapt quickly and may face the risk of extinction. Forests, wetlands, grasslands, and marine ecosystems are all vulnerable to climate change.

Human health is also at risk. Rising temperatures can increase heat strokes, dehydration, respiratory diseases, and heart-related problems. Air pollution caused by fossil fuel combustion can worsen asthma and other breathing disorders. Climate change may also increase the spread of certain diseases by changing the habitats of insects and disease-carrying organisms.

Greenhouse gas emissions also affect agriculture. Changes in temperature, rainfall, drought frequency, and extreme weather events can reduce crop production. This may lead to food shortages, higher food prices, and economic stress for farmers and consumers.

The economic effects are also significant. Extreme weather events such as floods, droughts, storms, and heatwaves cause damage to infrastructure, agriculture, industries, and homes. Governments and communities have to spend large amounts of money on disaster management, healthcare, repair work, and adaptation measures.

Measurement and Monitoring of Greenhouse Gas Emissions

Measuring and monitoring greenhouse gas emissions is important for understanding their sources and controlling their growth. Scientists use direct measurement, remote sensing, satellite observations, and computer-based analysis to track emissions.

Satellite observations provide a global view of greenhouse gas concentrations in the atmosphere. They help scientists monitor carbon dioxide, methane, and other gases across different regions.

Ocean and soil monitoring are also important because oceans and soils absorb large amounts of carbon. By studying carbon uptake in oceans, forests, and soils, scientists can understand how natural systems help regulate greenhouse gas levels.

Urban air quality monitoring is used in cities to measure pollutants and emissions from vehicles, industries, and buildings. Sensors placed in urban areas help identify pollution hotspots and support better environmental planning.

Advanced technologies such as big data, artificial intelligence, and machine learning are increasingly used to analyze emission trends. These tools help predict future emissions and evaluate the effectiveness of emission reduction policies.

Strategies to Reduce Greenhouse Gas Emissions

Several strategies can be used to reduce greenhouse gas emissions. One of the most important strategies is the adoption of renewable energy. Shifting from fossil fuels to cleaner sources such as solar, wind, hydropower, and geothermal energy can greatly reduce carbon dioxide emissions.

Energy efficiency is another effective method. Using energy-efficient appliances, machines, buildings, lighting systems, and industrial processes can reduce energy consumption and lower emissions.

Carbon capture and storage is a technology that captures carbon dioxide emissions before they enter the atmosphere. The captured carbon dioxide is then stored underground or used in industrial processes. This method can help reduce emissions from power plants and heavy industries.

Strong emission laws and environmental regulations are also necessary. Governments can set limits on industrial emissions, vehicle emissions, and pollution from power plants. Strict standards encourage industries to adopt cleaner technologies.

Financial incentives can also support emission reduction. Tax benefits, subsidies, and grants can encourage businesses and individuals to invest in renewable energy, electric vehicles, energy-efficient buildings, and clean technologies.

Research and development are important for finding new solutions. Investment in clean energy, battery storage, carbon capture, green hydrogen, and sustainable agriculture can help reduce emissions in the long term.

Public awareness is equally important. People should understand their carbon footprint and adopt simple actions such as saving electricity, using public transport, reducing waste, planting trees, and supporting sustainable products.

Policies and Regulations for Reducing Emissions

Policies and regulations play an important role in controlling greenhouse gas emissions. International agreements encourage countries to work together to reduce emissions and fight climate change. One important example is the Paris Agreement, under which countries aim to limit global warming and reduce greenhouse gas emissions.

National and regional governments also introduce policies to reduce emissions. These may include renewable energy targets, pollution control rules, energy efficiency standards, and climate action plans.

Corporate responsibility is also important. Many companies are setting emission reduction targets, improving energy efficiency, using renewable energy, and reporting their environmental impact.

Carbon pricing is another policy tool. It puts a price on carbon emissions through carbon taxes or cap-and-trade systems. This encourages industries to reduce emissions because pollution becomes more costly.

Green building standards help reduce emissions from the construction and housing sector. Energy-efficient buildings use less electricity for heating, cooling, and lighting, thereby reducing greenhouse gas emissions.

Transport rules can also reduce emissions. Governments can introduce stricter vehicle emission standards, promote electric vehicles, improve public transport, and encourage the use of cleaner fuels.

Support for climate research is necessary to improve understanding of climate change and develop better solutions for mitigation and adaptation.